AI RESEARCH ROADMAP · STAGE 1 · STEP 3

Multivariable Basics Complete Notebook

Where Linear Algebra and Calculus meet. Two topics only — but they are the two that explain how every machine learning model actually trains.

What is inside

2 topics	Partial derivatives → the gradient
Full lessons	Plain-language explanation of every concept
Diagrams	Contour maps showing gradient descent in action
Worked examples	Every calculation written out step by step
Tests + answers	3 questions per topic, easy · normal · hard
Master reference	All key points in one table

Only two units of the multivariable course are required. Multiple integrals, divergence and curl are skipped — not needed for machine learning.

Phase 1 · Stage 1 · Step 3

Partial derivatives

One derivative per input — the only new skill is freezing variables.

THE IDEA

Every function so far had one input. Real functions have many: a loss function depends on every weight in the network. So instead of a single derivative, you get **one derivative per input**, and each is called a **partial derivative**.

Symbol	Means	How to compute it
$\partial f/\partial x$	How f changes when only x moves	Treat y as an ordinary number
$\partial f/\partial y$	How f changes when only y moves	Treat x as an ordinary number

> **The only new skill: freeze the other variable.** The symbol ∂ is just "d" for partial — all the rules from Calculus (power rule, chain rule, product rule) apply completely unchanged.

TWO THINGS THAT TRIP PEOPLE UP

A term with **no x in it at all** is a constant when differentiating with respect to x, so it vanishes. And a term like x^2y is not a product needing the product rule — with y frozen, y is simply a multiplier sitting in front of x^2.

WORKED EXAMPLE

PARTIAL DERIVATIVES — STEP BY STEP

a function with TWO inputs instead of one

THE IDEA

$f(x, y)$ depends on two things. so there are TWO derivatives :

$\partial f/\partial x$ = how f changes when only x moves (treat y as a NUMBER)

$\partial f/\partial y$ = how f changes when only y moves (treat x as a NUMBER)

the symbol ∂ is just "d" for partial. the rules are the same rules.

EXAMPLE $f(x, y) = x^2y + 3y^2$

STEP 1 — find $\partial f/\partial x$ (y is frozen, like a constant)

$x^2y \rightarrow y \cdot 2x = 2xy$

y is just a multiplier sitting in front of x^2

$3y^2 \rightarrow 0$

no x in it at all → it is a constant → vanishes

$$\partial f/\partial x = 2xy$$

STEP 2 — find $\partial f/\partial y$ (now x is frozen)

$x^2y \rightarrow x^2 \cdot 1 = x^2$

x^2 is the multiplier, y differentiates to 1

$3y^2 \rightarrow 3 \cdot 2y = 6y$

$$\partial f/\partial y = x^2 + 6y$$

THE ONLY NEW SKILL : freeze the other variable and treat it as an ordinary number. Everything else is chapter 3 and 4.

> **Why it matters for ML:** a model with a million weights has a million partial derivatives. Each one answers the same question — "if I nudge this one weight, how much does the loss change?" — and that is exactly what backpropagation computes.

KEY POINTS FROM THIS TOPIC

Partial derivative	one derivative per input variable
∂f/∂x	freeze y, treat it as a number
∂f/∂y	freeze x, treat it as a number
Terms without that variable	become constants → differentiate to 0
Rules used	exactly the same as single-variable calculus

TEST — Topic 1

1. $f(x, y) = 3x + 4y$. Find $\partial f/\partial x$ and $\partial f/\partial y$. → **3 and 4**
2. $f(x, y) = x^2y^3$. Find both partial derivatives. → **$\partial f/\partial x = 2xy^3$ · $\partial f/\partial y = 3x^2y^2$**
3. $f(x, y) = x^2y + 5y$. Find both partial derivatives. → **$\partial f/\partial x = 2xy$ · $\partial f/\partial y = x^2 + 5$**

The gradient

Where linear algebra and calculus meet — and how models train.

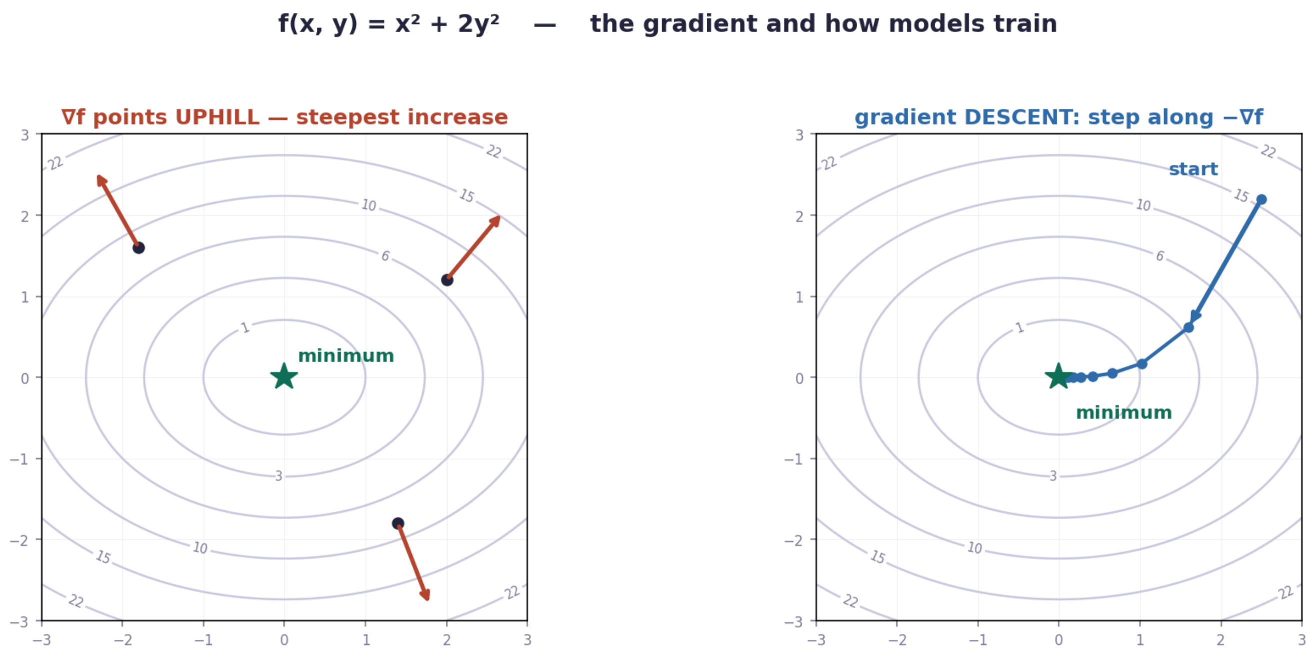
THE IDEA

Collect all the partial derivatives into a single **vector**. That vector is the **gradient**, written ∇f (say "grad f" or "del f").

> **$\nabla f = (\partial f/\partial x , \partial f/\partial y)$** — a vector whose components are the partial derivatives.

WHAT IT TELLS YOU

Property	Meaning
Direction	Points the way of **steepest increase**
Length	How steep that increase is
$-\nabla f$	Points the way of steepest **decrease** — downhill
$\nabla f = 0$	Flat: a minimum, a maximum, or a saddle point



EVALUATING A GRADIENT

The gradient is a formula until you plug in a point. Substituting a specific (x, y) turns it into an ordinary vector of numbers — exactly the kind you handled in Linear Algebra.

WORKED EXAMPLE

THE GRADIENT — ∇f

where linear algebra and calculus meet

THE DEFINITION

$$\nabla f = (\partial f/\partial x , \partial f/\partial y)$$

a VECTOR whose components are the partial derivatives

WHAT IT MEANS

direction	points the way of STEEPEST INCREASE
length	how steep that increase is
negative $-\nabla f$	points the way of steepest DECREASE

EXAMPLE $f(x, y) = x^2y + 3y^2$

from the previous page :

$\partial f/\partial x = 2xy$ $\quad$ $\partial f/\partial y = x^2 + 6y$

$$\nabla f = (2xy , x^2 + 6y)$$

EVALUATE AT A POINT $(x, y) = (1, 2)$

$\partial f/\partial x = 2(1)(2) = 4$

$\partial f/\partial y = (1)^2 + 6(2) = 13$

$$\nabla f\,(1,2) = (4 , 13)$$

a plain vector — exactly like week one

GRADIENT DESCENT : $w\_new = w\_old - learning\_rate \cdot \nabla f$
an anchor point, a direction, and a dial — the parametric line from Linear Algebra chapter 2. This is how models train.

Gradient descent — how every model trains: w_new = w_old – learning_rate · ∇f
An anchor point, a direction, and a dial. That is the parametric line from Linear Algebra Chapter 2, run in a space of millions of dimensions. The minus sign is what makes it descend rather than climb, and the learning rate is how big each step is.

KEY POINTS FROM THIS TOPIC

Gradient	$\nabla f = (\partial f/\partial x , \partial f/\partial y)$
Direction	steepest increase
$-\nabla f$	steepest decrease — downhill
$\nabla f = 0$	flat point: minimum, maximum or saddle
Gradient descent	w_new = w_old – learning_rate · ∇f

TEST — Topic 2

1. $f(x, y) = x^2 + y^2$. Find ∇f. **→ $(2x, 2y)$**
2. $f(x, y) = x^2y + 5y$. Find ∇f, then evaluate it at $(2, 1)$. **→ $\nabla f = (2xy, x^2 + 5)$; at $(2,1) = (4, 9)$**
3. $f(x, y) = 3x^2y$. Find ∇f and evaluate at $(1, 2)$. **→ $\nabla f = (6xy, 3x^2)$; at $(1,2) = (12, 3)$**

REFERENCE

Master reference sheet

Topic	Item	Rule
T1	**Partial derivative**	one derivative per input variable
T1	**∂f/∂x**	freeze y, treat it as a number
T1	**∂f/∂y**	freeze x, treat it as a number
T1	**Terms without that variable**	become constants → differentiate to 0
T1	**Rules used**	exactly the same as single-variable calculus
T2	**Gradient**	$\nabla f = (\partial f/\partial x , \partial f/\partial y)$
T2	**Direction**	steepest increase
T2	**$-\nabla f$**	steepest decrease — downhill
T2	**$\nabla f = 0$**	flat point: minimum, maximum or saddle
T2	**Gradient descent**	w_new = w_old − learning_rate · ∇f

VERIFY HABITS

- **Partial derivatives:** check every term without that variable became 0
- **Freezing:** the frozen variable stays exactly as it is — never differentiate it
- **Gradient:** it must have one component per input variable
- **Evaluating:** substitute the point into every component, not just the first
- **Sign:** descent uses **minus** the gradient — plus would climb the hill

PERSONAL ERROR LOG — CARRIED FORWARD

Error	Fix
Misreading given numbers	Copy all values onto paper, labelled, before computing
Dropping a minus sign	Write subtraction as adding a negative
Assuming a shortcut applies	Check the condition first
Rushing the easy question	Same care on all three questions